

SUPPLEMENTARY MATERIAL

*TO BE SUBMITTED TO:
THE JOURNAL OF CHEMICAL PHYSICS

Glassy Dynamics in Mutant Huntingtin Proteins

Hongsuk Kang, Binquan Luan, and Ruhong Zhou*

*Computational Biology Center, IBM Thomas J. Watson Research Center, Yorktown Heights, NY, 10598,
USA*

*Authors to whom correspondence should be addressed. Electronic addresses:
ruhongz@us.ibm.com*

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Simulated temperatures

Temperature range used in our simulation is presented in Table S1. The lowest temperature is 295.89K and the highest is 645.56K.

INDEX	TEMP	INDEX	TEMP	INDEX	TEMP	INDEX	TEMP	INDEX	TEMP
1	295.89	27	339.49	53	394.02	79	461.15	105	548.33
2	297.38	28	341.38	54	396.35	80	464.10	106	552.14
3	298.88	29	343.29	55	398.69	81	467.07	107	555.99
4	300.39	30	345.20	56	401.05	82	470.08	108	559.88
5	301.92	31	347.15	57	403.43	83	473.12	109	563.80
6	303.46	32	349.10	58	405.83	84	476.20	110	567.76
7	305.02	33	351.07	59	408.24	85	479.30	111	571.75
8	306.59	34	353.06	60	410.67	86	482.43	112	575.79
9	308.19	35	355.06	61	413.11	87	485.60	113	579.86
10	309.79	36	357.09	62	415.59	88	488.80	114	583.97
11	311.41	37	359.13	63	418.07	89	492.03	115	588.12
12	313.05	38	361.18	64	420.57	90	495.30	116	592.30
13	314.70	39	363.25	65	423.07	91	498.60	117	596.52
14	316.37	40	365.34	66	425.60	92	501.93	118	600.78
15	318.05	41	367.45	67	428.16	93	505.29	119	605.09
16	319.75	42	369.57	68	430.74	94	508.69	120	609.42
17	321.47	43	371.71	69	433.36	95	512.12	121	613.80
18	323.19	44	373.86	70	436.01	96	515.58	122	618.22
19	324.94	45	376.03	71	438.68	97	519.08	123	622.67
20	326.70	46	378.21	72	441.38	98	522.62	124	627.17
21	328.48	47	380.42	73	444.12	99	526.19	125	631.70
22	330.28	48	382.65	74	446.88	100	529.79	126	636.29
23	332.09	49	384.88	75	449.67	101	533.43	127	640.90
24	333.92	50	387.14	76	452.50	102	537.10	128	645.56
25	335.76	51	389.41	77	455.35	103	540.80		
26	337.61	52	391.71	78	458.24	104	544.55		

Table S1 Temperatures used in T-REMD simulations. The unit is Kelvin (K). The temperature ranges from 295.89K to 645.56K.

Representative structure for Q56

We selected the representative structures for Q56 at 298K from T-REMD simulations using cluster analysis and used as initial structures for unbiased MD over all temperatures. GROMOS¹ algorithm with 3.5Å cut-off for C_{α} atoms was used to choose top five clusters. The results are presented in Figure S1.

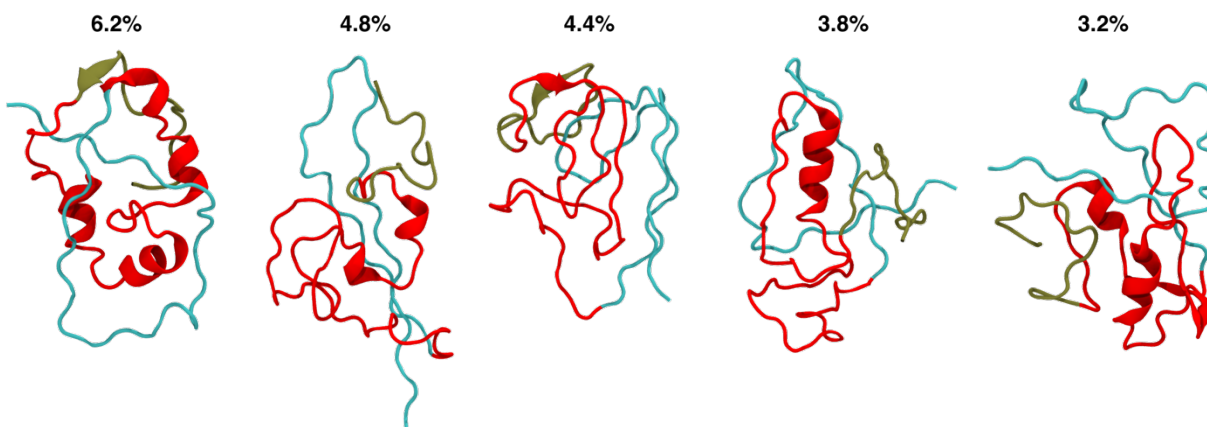


Figure S1 The representative structures for Q56 at 298K chosen by cluster analysis with GROMOS algorithm in New Cartoon Representation. The cut-off distance was 3.5Å. N17, polyQ and C38 are colored by tan, red and cyan, respectively. The number above each structure indicates the percentage of each cluster occupying the whole conformational space.

Mean-squared displacement of T-REMD data

We compared the mean-squared displacement (msd) of T-REMD data with the msd of unbiased MD (Figure S2). Interestingly, the msd of T-REMD is between the msd of 480K and 600K data, implying that the replica exchange method relaxes the system with efficiency.

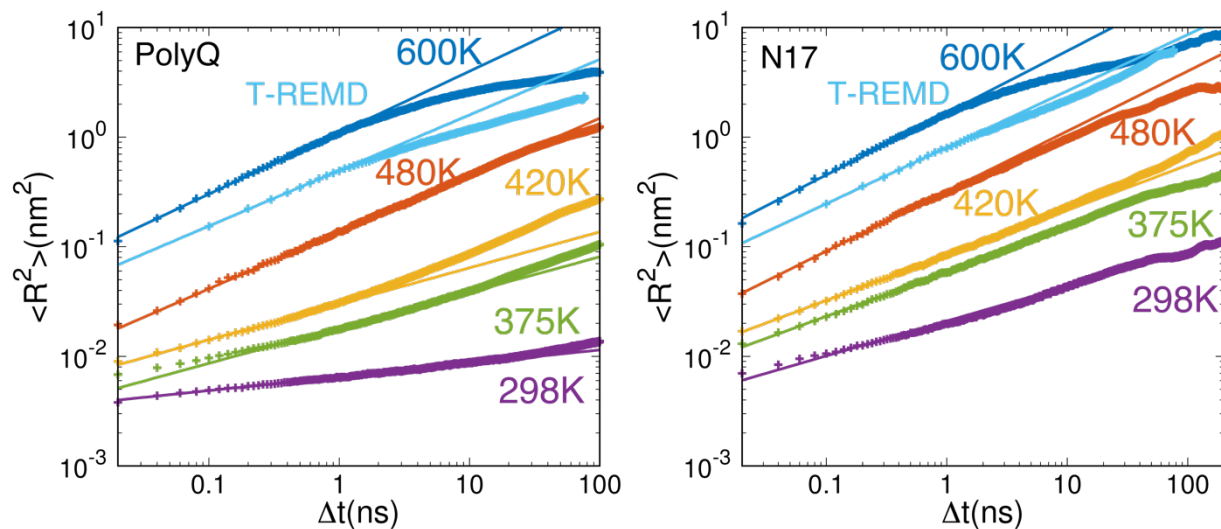


Figure S2 The comparison of the mean-squared displacement (msd) of polyQ (left) and N17 (right) for T-REMD (cyan) and unbiased MD (blue, red, orange, green and purple) simulations. As described in main text, the msd was calculated for C_{α} atoms after alignments. The msd of T-REMD is between ones for unbiased MD simulations at 600K and 480K, indicating accelerated dynamics by replica exchange.

Secondary structure over 300ns for 298K

Timeseries of secondary structure content for five different runs (Figure S3). The plot shows little change as a function of time until 300ns.

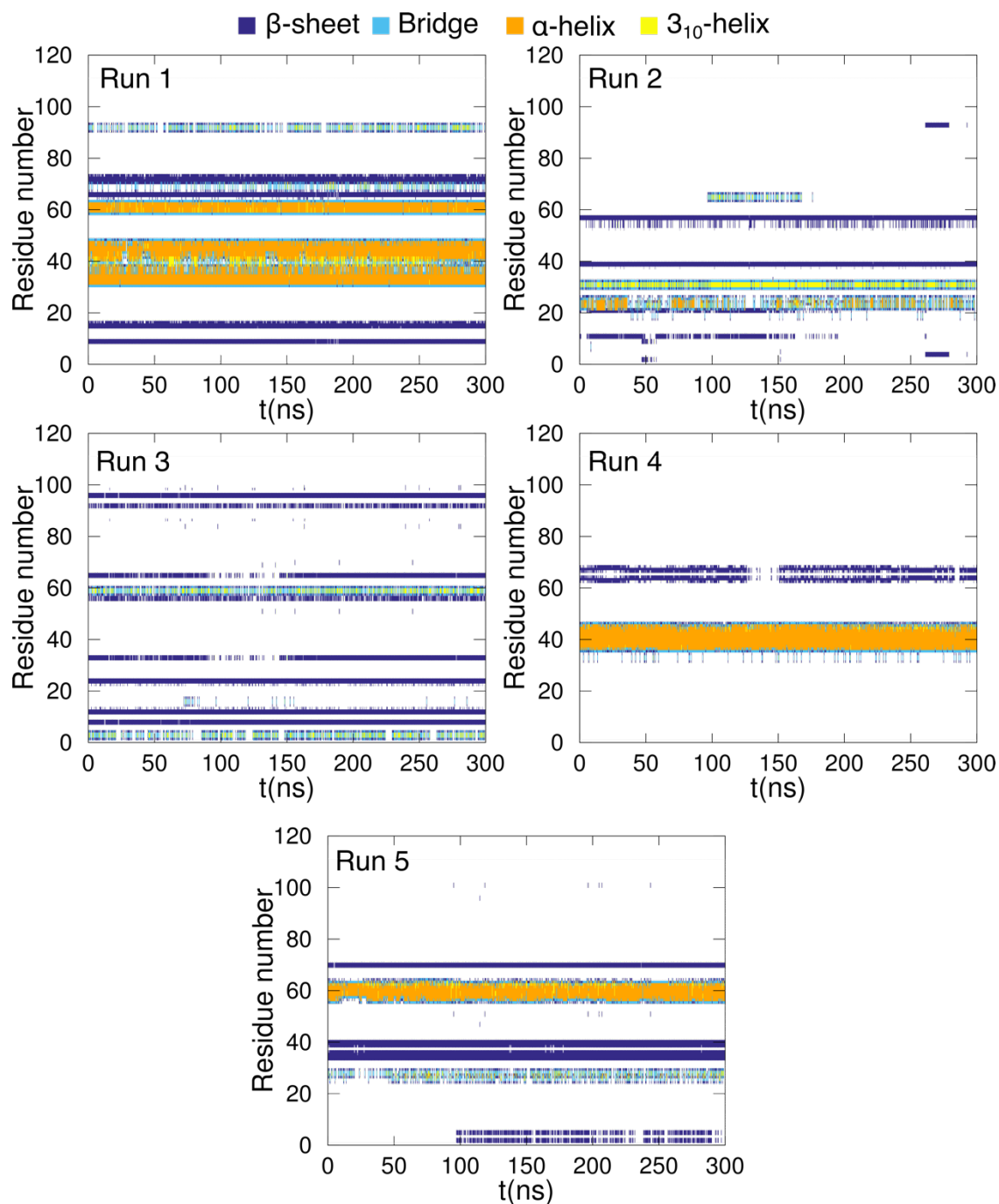


Figure S3 The timeseries of secondary structure for five different runs of unbiased MD with Q56 at 298K. Secondary structures are stable over 300ns except minor changes due to thermal fluctuations.

References

¹ X. Daura, K. Gademann, B. Jaun, D. Seebach, W. F. van Gunsteren, and A. E. Mark, *Angew. Chem. Int. Ed.* **38** (1999) 236.